Unit 7 "Metrics for Experimentation for Experimentation Criterion".

1) From Business Metrics to Metrics Appropriate for Experimentation:

(1) Why not use Raw Business Metrics in Experiments?

Business metric like:

- Total Revenue
- Quarterly Profit
- Net Promoter Score.

... are important, but they're not usable directly in short-term A/B tests.

X Problem 1: Not Observable Quickly

Business metrics are too slow. You can't wait 3 months to know if your new button works.

⇒ A/B testing needs metrics that change fast - within days or weeks.

X Problem 2: Too Noisy / High Variance

Metrics like revenue are very noisy. Why?

- Most users buy nothing
- A few users (big spenders) dominate the total
- This makes it hard to detect small improvements

As a result:

Even good changes may show no statistical significance

(2) What Makes a Metric Good for Experimentation?

we want a metric that is:

Property	Why it Matters.
Sensitive	Detects small changes quickly (→ more power)
Timely	Measurable during the experiment window.
Directional	Improvements always mean positive impact.
Low variance	Keep noise down → clear signal.
Hard to game	Resist manipulation by users or devs.

(3) Real-world Analogy: A Coffee Shop.

Imagine you run a coffee shop and try a new menu.

The CEO cares about:
- "Did profit increase this quarter?"
  But this is too slow to wait for.

You use this instead:
- Average number of items bought per customer per day.

Why?
- Measurable quickly
- Lower noise
- Predicts the final business goal

This is like tasting the soup while it's cooking, instead of waiting for everyone to finish eating.

(4) The math behind the transformation.

Let

$$Y_{u,i} = \text{outcome for user } u \text{ in variant } i \in \{C, T\}$$

A classic average-treatment-effect estimate is

$$\Delta = \bar{Y}_T - \bar{Y}_C, \quad SE(\Delta) = \sqrt{\frac{s_T^2}{n_T} + \frac{s_C^2}{n_C}}$$

If we redefine Y_{ui} from total revenue to a lower-variance proxy (e.g. conversion indicator), s_p^2 plummets, shrinking the standard error and boosting power. The theorem is simple: for fixed effect size, reducing variance is mathematically identical to adding users or test days.

Example:

Let's say:

- You're testing a feature that increases purchases from 10% to 12%
- You measure revenue per user vs conversion rate.

Metric	Avg. Variance	Power (chance of detecting change)
Revenue	475.4	23.8% ✗ Very low
Conversion rate	0.0899	89.0% ☑ Better.

Goal: Estimate the Treatment Effect

Let:

- μ_C: Mean of control group.
- μ_T: mean of treatment group.
- s_C^2, s_T^2: variance of control and treatment.
- n_C, n_T: sample sizes of each group.

Then the estimate effect (diff) is

$$\Delta = \mu_T - \mu_C$$

The standard Error (SE)

This tells how noisy your effect estimate is

$$SE(\Delta) = \sqrt{\frac{s_C^2}{n_C} + \frac{s_T^2}{n_T}}$$

$\Rightarrow$ lower variance s^2 = lower SE = better precision.

95% Confidence Interval

This gives a range in which the true effect likely lies (with 95% certainty):

$$\Delta \pm 1.96 \cdot SE$$

$$CI = \left[\mu_T - \mu_C - 1.96 \cdot \sqrt{\frac{s_C^2}{n_C} + \frac{s_T^2}{n_T}} \;,\; \mu_T - \mu_C + 1.96 \cdot \sqrt{\frac{s_C^2}{n_C} + \frac{s_T^2}{n_T}} \right]$$

Key insights

Suppose you hold:

- $n_C \approx n_T \approx n$
- $s_C \approx s_T \approx s$

Then:

$$SE \approx \sqrt{\frac{s^2}{n} + \frac{s^2}{n}} \approx \frac{s\sqrt{2}}{\sqrt{n}} \approx SE_{base}.$$

> Halves variance = double the sample size

$$SE_A \approx SE_B.$$

$$\sqrt{\frac{s^2/2}{n} + \frac{s^2/2}{n}} \approx \sqrt{\frac{s^2}{2n} + \frac{s^2}{2n}}$$

$$\frac{s}{\sqrt{n}} \approx \frac{s}{\sqrt{n}}$$

Conclusions

If you're experimenting with:

- High-variance metric: → wide confidence interval, harder to detect effects.
- Low-variance metric → tight interval, higher chance of significance.

(5) A/B Test Variance Analysis : Why Metric choice Determines Success.

(1) The Core Problem

When A/B testing the same feature improvement (10% → 11% conversion lift), different metrics yield dramatically different statistical power.

Metric	Variance	Standard Error	Statistical Power
Conversion	0.0899	0.006	39.0%
Revenue	475.4	0.406	23.8 %

(2) Mathematical Deep Dive:

Option A : Binary Conversion Rate

For a binary metric like where $p = 0.10$ (10% conversion):

- Mean (μ) $= 0.10$
- Variance (σ^2) $= p(1-p) = 0.10 * 0.90 = 0.09$
- Stand Error : $\sqrt{(0.09/5000 + 0.098/5000)} = 0.006$

2-score calculation

Effect size (Δ) $= 0.11 - 0.10 = 0.01$

$z = 0.01 / 0.006 \approx 1.67$

p-value ≈ 0.095 (close to significance)

Option B : Revenue per User (Heavy-tailed Distribution)

For revenue with high variance

- Mean : $10
- Variance : 475 (extremely high due to skewness

- Standard Error : $\sqrt{(475/5000 + 475/5000)} \approx 0.436$.

z-score for $1 ~~revenue~~ increase :

Effect size (Δ) = \$11 − \$10 = \$1

$z = 1 / 0.436 \approx 2.29$ (seems good).

But for realistic small effects (\$0.25)

$z = 0.25 / 0.436 \approx 0.57$

p-value >> 0.05 (not significant).

③ The Variance – Power Relationship

Fundamental Formula

Statistical Power ∝ (Effect Size) / (Standard Error)

Standard Error = $\sqrt{(\text{variance} / \text{sample size})}$

Therefore:

Power ∝ Effect size / $\sqrt{(\text{variance}/n)}$

Power ∝ (Effect size × $\sqrt{n}$) / $\sqrt{\text{variance}}$

Key Insights

Reducing variance by half has the same impact on power as doubling your sample size.

④ Why High-Variance Metrics Fails

- Typical e-commerce : 80-90% of users spend \$0
- Power users : 5-10% generate 50-80% of revenue.
- Whales : 1-2% might spend 10× the average.

This creates:

- High variance due to extreme outliers
- Skew distribution that violate normal assumptions
- Unstable estimates that require massive sample sizes.

Variance Comparison

Conversion Rate Variance : 0.09 (bounded by definition)
Revenue Variance : 475+ (unbounded, heavy tailed)
Ratio: 475 / 0.09 ≈ 5,278 × higher.

Why High-Variance Metrics fails

Revenue / Distribution / Reality

Practical × Metric Selection strategy.

☑ Low-variance Metrics (Prepared)

1. Conversion rates (0/1 bounded)
2. Clicks-through-rates (0/1 bounded)
3. Sessions completion rates (0/1 bounded)
4. Average order value (conditional on purchase)
5. Log-transformed revenue (reduces skewness)

☒ High-Variance Metrics (Avoid for Primary Analysis)

1. Raw revenue per user
2. Total profit
3. Lifetime value (early estimate)
4. Session duration (heavy-tailed)
5. Page views per users.

Advanced Variance Reduction Techniques.

1. Ratio Metrics

Instead of raw revenue, use:

- Revenue per session (normalizes by engagement)
- Revenue per click (normalizes by interest)
- Conversion rate (ultimate normalization).

2. Transformation Methods.

- Log transformation: log(revenue + 1) for skewed data
- Winsorization: Cap extreme values at 99th percentile
- Trimming: Remove top/bottom 1% of observations.

3. Stratification

Segment users by:

- Historical spending (group similar users)
- Geography (different purchasing power)
- Device type (mobile vs desktop behavior)

4. CUPED (Controlled-Experiment Using Pre-Experiment Data)

Adjusted Metric = Raw Metric − θ × (Pre-period Metric − Pre-period Mean).

This can reduce variance by 20–50% using historical data.

Power Analysis Formula:

(1)

The Power Analysis formula helps you answer a critical question before running any experiment.

> "How many users do I need in each group (control / treatment) to detect a meaningful effect with high confidence?"

This is essential to avoid two major pitfalls:

Pitfalls	without Power Analysis
X Underpowered Test	You might miss a real effect (false rejecting) because the sample is too small.
X Overpowered Test	You might waste traffic/time/ resources running a longer than necessary test.

What Power Analysis Ensures:

You'll detect a meaningful change (Δ) with...

- Statistical confidence (e.g. 95%)
- Statistical power (e.g. 80%)
- Given your metric's variance (σ^2)

Formula:

$$n = 2 \cdot (Z_{\alpha/2} + Z_{\beta})^2 \frac{\sigma^2}{\Delta}$$

where,

Symbol	Meaning	Typical value
$z_{\alpha/2}$	Z-score for 95% confidence	1.96
z_β	Z-score for 80% power	0.84
σ^2	Variance of your metric	← depends on metric
Δ	Minimal effect you want to detect (MDE)	set by product goals.

(2)

Example: Conversion Rate (Low Variance Metric)

Let's say

- Baseline conversion rate: 10% ($p = 0.10$)
- Minimum effect to detect (Δ): 1% absolute lift (0.01).
- Variance for binary metric

$$\sigma^2 = p(1-p) = 0.10 \cdot 0.90 = 0.09$$

Plug into formula:

$$n = 2 \cdot (1.96 + 0.84)^2 \cdot \frac{0.09}{0.01^2}$$

$$= 2 \cdot (2.80)^2 \cdot \frac{0.09}{0.0001}$$

$$n = 2 \cdot 7.84 \cdot 900 = 14,112 \text{ per group.}$$

✓ You only need ~ 14,000 users per group to detect a 1% lift in conversion.

Example 2: Revenue per user (High Variance Metric)

Let's say:

- Mean revenue per user = $10
- But variance = 475 (because most users spend $0, some spend $500)
- You still want to detect a $1 lift ($\Delta = 1$)

Plug into formula:

$$n = 2 \cdot (1.96 + 0.84)^2 \cdot \frac{475}{1}$$

$$n = 2 \cdot 7.84 \cdot 475 = 7,448 \cdot 2 = 14,896$$

$$= 14,896 \text{ per group}$$

⚠ This looks similar to conversion, but now...

What if you want to detect a $0.25 lift?

$$n = 2 \cdot (1.96 + 0.84)^2 \cdot \frac{475}{0.25^2} = 2 \cdot 7.84 \cdot \frac{475}{0.0625}$$

$$= 2 \cdot 7.84 \cdot 7,600 = 119,168 \text{ per group}$$

Now you need ~120k users per group, just because variance is high.

Conclusion: Revenue metric needs 8x more traffic to detect the same-sized effect.

(3) Summary of Lessons

☑ Use Low-Variance Metrics for Primary Decisions

- Conversion Rate
- Click-Through Rate
- Log-transformed revenue
- Revenue per session, not total

✗ Avoid high-variance Metrics for Core Evaluation

- Raw Revenue per User.
- Total Revenue or Profit
- Time-on-site or Page Views (heavy-tailed).

(4) Takeaway formulas to Remember

$$n \propto \frac{\sigma^2}{\Delta^2}$$

(1) smaller effects Need Larger Sample Size

↳ Δ is small $\Rightarrow \Delta^2$ is very small

↳ that makes $\frac{\sigma^2}{\Delta^2}$ big $\Rightarrow$ n increases.

So: | smaller the improvement you care about → more users you need.

(2) Lower Variance Means Fewer Users Needed.

If your metric has less noise (lower σ^2):

- $\frac{\sigma^2}{\Delta^2}$ becomes smaller $\Rightarrow$ n decreases.

So: | cleaner, more stable metrics (like conversion rate) need less traffic.

(6) Connection to Other Topics

- Combining Metrics into OEC: when a single metric isn't enough, we combine several (e.g. conversion rate + time on site).
- Goodhart's law: "when a metric becomes a target, it ceases to be a good metric" → You must choose metrics that don't get gamed.
- CUPED: a statistical trick that reduces variance without changing your metric, by using pre-experiment behavior as a covariate.

(7) Summary – How to convert Business Matric to Experiment Matrics.

Bussiness Matric	Problem	Experiment Matric Example.
Total revenue	High variance	Revenue per user
Quarterly profit	- Too slow	Daily gross margin per customer.
NPS survey	Too sparse	Satisfaction % from quick surveys.

(2) Combining key Metrics into an OEC

(1) Why do we need an OEC?

→ In A/B testing, multiple metrics often move at once.

- ↳ Conversion rate ↑
- ↳ Revenue per user ↓
- ↳ Engagement ↑

Question: Which variant is better?

You can't decide unless you:

- Combine these into a single number that reflects overall product value.

→ That's the OEC: Overall Evaluation Criterion.

(2) What is an OEC?

→ OEC: a weighted combination of key metrics so you can make one clear decision.

Math form:

$$OEC = w_1 \cdot M_1 + w_2 \cdot M_2 + \dots + w_k \cdot M_k$$

- $M_1, M_2 \dots$ are your core metrics (e.g. conversion, revenue)
- $w_1, w_2 \dots$ are weights (summing to 1) showing how much each metric matters.

Building blocks

Step	What you do	Math view
Select 2–6 core metrics	must cover revenue, engagement, and health (e.g. latency)	$M_1, \dots, M_k$

Normalise/scale	put everything on comparable ranges (divide by baseline, z-score etc)	$\tilde{M}_i$
Assign weight w_i	based on business value, elasticity studies, exec input	$w_i \geq 0$, $\sum w_i = 1$
Compute OEC	weighted sum (or geometric/utility)	$OEC = \sum w_i \tilde{M}_i$

(3) Real-life Analogy : School GPA:

Imagine a student:

- Math (weight: 0.5)
- English (weight: 0.3)
- Art (weight: 0.2)

final GPA = 0.5 · Math Score + 0.3 · English + 0.2 · Art.

Similarly, in A/B testing

- Conversion rate (weight: 0.6)
- Revenue per user (weight: 0.4)

OEC = 0.6 · conversion + 0.4 · normalized revenue

(4) with and without OEC

Suppose you run an A/B test, and you see results like these:

Metric	Control	Treatment	Change
✓ Conversion Rate	10%	11%	+10%
✗ Rev/user	$5.00	$4.80	-4%

| so, did the new version win or lose?

one manager says: "Conversion went up! ship it"
Another says: "Revenue dropped! Revert it!"
Analyst: "It's a trade-off, but which is more important?"

without OEC

- You debate endlessly
- Everyone defends their favorite metric
- Decisions get delayed
- You risk making emotional or biased choices.

with OEC (one combined score)

Now suppose you say:

| "Let's pre-define the importance of each metric. 60% weight on Conversion Rate, 40% on Revenue per User."

Then you calculate a single score (OEC)

OEC = 0.6 · conversion + 0.4 · normalized revenue

Now if:

- OEC ↑ → The treatment is better overall → ship ✓
- OEC ↓ → Treatment worse overall → Reject ✗

⇒ OEC gives you a single, fair, trusted number to decide if an experiment worked overall, even when some metrics go up and other go down.

(5) Tips and warning

Do	Don't
☑ set weights before experiment starts	☒ Adjust weights after seeing results ("metric
☑ Include "guardrail metrics" (like latency or bounce rate)	☒ Combine everything blindly
☑ Use low-variance versions (per-user normalized)	☒ Use raw variance or total directly.

(6) Links to other topics.

- From Business Metrics → Experiment Metrics: OEC relies on experiment ready building block (fast, low-variance)
- Goodhart's / Campbell's law: Once OEC is public, teams may game sub-metrics; periodically re-validate the weights.
- Amazon e-mail OEC, Bing search OEC: upcoming pages show concrete weightings; Amazon & Microsoft choose and how they evolved.

(7) Key takeaways

An OEC is a contract
Agree on the recipe before the experiment, keep it stable, and you get faster, less-political product decisions.

③ Example: OEC for E-mail at Amazon

① How Amazon Designed an OEC for Marketing E-mails.

"The Problem Amazon Faced"

→ Amazon used to evaluate email experiments like this:

> "Did the email generate more revenue?"

But teams started abusing this:

- More emails = more clicks = more revenue in short term.
- But many customers got annoyed and clicked "Unsubscribe".
- Unsubscribed users = permanently lost future revenue

→ This made "spammy" experiments look good — even if they hurt Amazon's long-term business.

② Refix: A New OEC Formula

Amazon added a penalty for unsubscribes into their Overall Evaluation Criterion (OEC)

$$OEC = \frac{\Sigma \text{ Revenue} - L \cdot S}{n}$$

Where:

- Σ Revenue = total revenue from emails
- L = estimated lifetime value lost per unsubscribe (e.g. $12)
- S = number of unsubscribes.
- n = total users in the experiment.

User	Clicked Email	Revenue ($)	Unsubscribed?
A	✓ Yes	10.00	✗ No
B	✓ Yes	15.00	✗ No
C	✗ No	0.00	✓ Yes
D	✓ Yes	12.00	✗ No
E	✗ No	0.00	✓ Yes

- Total revenue

  R: 10 + 15 + 0 + 12 + 0 = 37

- Total users

  n = 5

- Number of unsubscribes

  s = 2

- Life time loss per unsubscribe (L)

  L = Avg Revenue (subscribed) - Avg Revenue (Unsubscribed)

  but, we assume

  L = 12

Compute OBC

$$OBC = \frac{R - L \cdot s}{n} = \frac{37 - 12 \cdot 2}{5} = \frac{37 - 24}{5} = \frac{13}{5} = 2.6$$

Interpretation

- The raw revenue per user was: $\frac{37}{5} = 7.4$
- But after subtracting the cost of unsubscribe, the OBC is only 2.6 per user.
- That means the campaign looks much worse when you consider long-term impact.

What if there were no unsubscribes?

Then:

$$S = 0 \Rightarrow OBC = \frac{37-0}{5} = 7.4$$

The OBC would match the revenue per user - ideal case.

What if we got 1 more unsubscribe?

Let's say User A also unsubscribed $\rightarrow S = 3$

$$OBC = \frac{37-12.3}{5} = \frac{37-36}{5} = \frac{1}{5} = 0.2$$

Now the campaign is barely breaking even: $0.20 per user.

Key Math Insight

- This OBC behaves like a linear function.

$$OBC(S) = \frac{R-12.S}{h}$$

It's strictly decreasing in S - each new unsubscribe hurt equally.

- if

$$S > \frac{R}{L} \Rightarrow OBC < 0$$

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$$CLC = \frac{R - L \cdot s}{n} = \frac{37 - 12 \cdot 2}{5} = \frac{37 - 24}{5} = \frac{13}{5} = 2.6$$

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② Why it Matters:

Old Metric (Revenue Only) → encourages short-term wins

New OEC (Revenue - Unsub Penalty) → align with long-term business health.

④ Related Concepts:

Related Concept	Descriptions
Combining Metrics into an OEC	This is real-world use of weighted metrics: revenue = +, unsubscribe = −
Goodhart's Law	when a metric becomes a target, it stops being good measure. Revenue alone was gamed.
Campbell's Law	Measuring performance using ↑ variables invites corruption. Fix: add more context like unsub rate.
Bing's OEC	Bing faced similar tension with short-term ad clicks vs long term user loyalty.

⑤ Final Takeaways:

An OEC must balance short-term profit with long-term trust. By subtracting loss for each unsubscriber, Amazon's OEC rewards helpful, valuable emails and penalizes annoying ones.

④ OEC for Bing's Search Engine.

① The Problem Bing Faced.

→ Bing ~~runs~~ runs thousands of A/B experiment to improve it's search engine. Each experiment could:

- Improve user satisfaction (e.g. better search results)
- Increase ad revenue (e.g. show more ads)

But these goals often conflict:

More ads = more money
But more ads = less satisfaction (users leave, bounce or churn).

So just optimizing revenue is dangerous.

② The solution: An OEC that balances Both.

→ Microsoft defined an Overall Evaluation Criterion (OEC) like this

$$OEC = \alpha \cdot \text{Success Rate} + \beta \cdot \text{Normalized Ad Revenue}$$

where:

- Success Rate: % of searches that were "satisfying" (e.g. a click + long dwell time)
- Ad Revenue: normalized per-search revenue (so different tests can be compared fairly).
- weights typically $\alpha = 0.7$, $\beta = 0.3$.

This single metric aligns both:

- short-term (ads)
- long-term (user trust)

Example: A/B Test on Bing

Metric	Control (A)	Treatment (B)	Lift %
Sessions/Action Ratio	61.9%	59.9%	↑ 3.1%
Revenue per Search	$0.3200	$0.0855	-72.9%
OEC per Search	0.7334	0.7199	-1.85%

Interpretation:

- Treatment B showed more ads but made users slightly less satisfied.
- The OEC decreased, so Bing rejected the experiment.

(3) Related Concept

Concepts	Connections
Combining Metrics into OEC	This is a perfect example of weighted metric fusion
Goodhart's Law	If Bing only used revenue teams would game it by flooding ads.
Amazon Email OEC	Some idea: reward good short-term actions, penalize user-hurting behavior.
Statistical Power	Using one combined metric reduces false positive vs. many separate ones.

(4) Key Takeaways

Bing OEC balances user happiness (40%) and revenue 30%). Any experiment that annoys users too much, even if it earns more in the short term gets automatically rejected.

(5) Goodhart's Law, Campbell's Law, and the Lucas Critique.

1. Goodhart's Law

> "When a measure becomes a target, it ceases to be a good measure"

Simple Meaning:

If teams optimize a metric too hard, they game the system — and the metric no longer reflects the true goal.

Example from A/B Testing:

→ You use "time spent on page" as a proxy for engagement

→ A product manager adds a loading spinner that delays page interaction.

→ Time goes up — but users are frustrated, not engaged.

Math Analogy (from simulation above)

→ Imagine your metric = true quality + noise

→ Before targeting: strong correlation between metric and truth

→ After optimizing only for top scorers: correlation drops

↳ You're now selecting outliers and overfit noise.

(1) Real-World Analogy (Super Simple).

Again: Goodhart's Law states | "When a measure becomes a target, it stops being a good measure."

This means:

* A metric that used to measure success.
* Now becomes the thing everyone optimizes.
* But people find ways to inflate the number without

improving real outcomes.

Imagine a teacher being evaluated only by student exam scores.

* The teacher wants a raise.
* So they "optimize" for high scores.
  - ↳ Gives students the answer key in advance
  - ↳ Teaches only question from last year's exam.

| Exam scores go up ✓
| Actual student learning goes down ✗

The exam score used to measure learning.
Now it's been manipulated to hit a target.

This is Goodhart's Law.

(2) Math-Based Analogy

Let's say you want to measure true user satisfaction, but you can't observe it directly. So you use a proxy metric - for example, time spent on the site.
Let:

- true-quality : real satisfaction (latent / unobserved)
- metric = true-quality + random-noise

Before metric is a target:

- You observe the metric passively
- The correlation between metric and true-quality is high (like 0.8)
- You trust the metric to rank users or content.

After metric becomes a target

- You say, "Let's promote the top 20% by this metric".
- But you're now selecting not just the best, but also those who got lucky noise (like outliers)
- Now, the correlation between ~~before~~ metric and true-quality drops (e.g. 0.63 in simulation).

That drop means:

You are now selecting based on noise, not truth.
→ This metric is no longer trustworthy.

A/B Testing Example (Concrete)

Let's say you run an A/B test and use "clicks per session" as your main metric

Original idea:

- click = = user is exploring = good.

After targeting:

- Teams start adding fake buttons or trick designs that increase clicks
- Result:
  - L) clicks go up ✓
  - L) Users are confused or annoyed ✗.

You optimized the metric — but harmed the user experience.

Take aways

When you start rewarding people for a metric, they'll optimize the number, not the outcome.

So, metrics need:

- Multiple components (like an OEC :. engagement + satisfaction).
- Guardrails to prevent gaming
- Randomization to detect real causal effect.

(2) Campbell's Law

> "The more any metric is used to make important decisions, the more likely it is to be corrupted — and to corrupt the process ~~the~~ it is supposed to monitor.

Simple Meaning:
when a metric becomes high-stakes, people will:

- Cheat
- Manipulate data
- Distort the process.

--- to make the numbers look good.

(1) Simple A/B Testing Example (Broken Relation).

Suppose a company says:

> "If the A/B test shows a 5% improvement in 7-day retention, the team gets a bonus".

what might the team do?

- Exclude new users who are unlikely to come back (i.e. churn-prone users).
- Delay bugs or bad features until the test is over.
- Push fake notifications to keep users active artificially.

Result:

The metric improves on paper.

✗ But it doesn't reflect true product improvement

This is Campbell's law — the metric becomes corrupt, and the test process loses value.

(2) Real-life Analogy: Test Scores.

Imagine teachers are judged solely on their students' math exam scores.

Here's what may happen:

- They teach only test topics (skip broader learning)
- They give practice questions that are identical to the test.
- In extreme cases, they might even change answers or hint during the exam.

Result:

- ☑ Test scores go up.
- ☒ Actual math ability stays the same or even worsens.

That is Campbell's law — the indicator is no longer a true measure of the real outcome.

(3) Math-Based Example.

Let's say:

True Quality = Produced Value

But you're only measured on:

Metric = Reduction Rate.

Now you redefine your experiment to only include users likely to return. You're filtering out the users who would lower your score — this inflates your metric.

It looks like you're doing better (metric ↑) but in reality, you're gaming the system, not improving the product.

Let:

- Q_i^* : the true quality of the i-th user (e.g. how much long-term value they get from the product)
- R_i : a retention indicator, where:

$$R_i = \begin{cases} 1 & \text{if user returns in 7 days} \\ 0 & \text{otherwise} \end{cases}$$

- Define true retention metric over a population S of users:

$$M(S) = \frac{1}{|S|} \sum_{i \in S} R_i^*$$

Now we show how filtering the sample S (changing picking) inflates $M(S)$ without changing any individual behavior.

Pure Symbolic Case.

Let's say your total user population is:

$$S = \{1, 2, 3, 4, 5, 6\}.$$

Let's define their true retention behaviors (fixed ahead of time):

User i	True Quality Q_i^*	Returns (7 days) R_i^*
1	0.9	1
2	0.8	1
3	0.7	1

4	0.6	0
5	0.5	0
6	0.4	0

* Case A: Honest Reporting

use all users $S = \{1, 2, 3, 4, 5, 6\}$

$$M(S) = \frac{1}{6}(1+1+1+0+0+0) = \frac{3}{6} = 0.5$$

* Case B: Manipulated Reporting (campbell's Law)

Exclude churn-prone users $\{4, 5, 6\}$, and define new set:

$$S' = \{1, 2, 3\}$$

Then:

$$M(S') = \frac{1}{3}(1+1+1) = 1.0$$

Insight

You did not change any user behavior, but your metric went:

- from 0.5 (honest)
- To 1.0 (Manipulated)

By just removing users who would make you look bad. This is the core mathematical idea of Campbell's Law.

Generalized Expressions

Let:
- S = All users
- $C \subset S$ = "clean" subgroup (e.g. high-likelihood returners).
- You'll define your observed metric over C, not S.

True metric: $M(S) = \frac{1}{|S|} \sum_{i \in S} R_i$

Manipulated: $M(C) = \frac{1}{|C|} \sum_{i \in C} R_i$ where $C = \{i \in S \mid Q_i > Q_3\}$

You get higher M(c), not because R_i changed, but because you excluded the low retention part.

(3) How Campbell's Law differs from Goodhart's Law

Law	Core problem	Example result
Goodhart's	Optimizing metric degrades signal	Metric becomes weak proxy for true quality.
Campbell's	People cheat when metric has power	Test is rigged to look good, but result is fake.

⇒ Goodhart is about proxy failure.
⇒ Campbell is about manipulation or fraud.

(4) How to Defend Against Campbell's Law

Strategy	Why it helps.
Use multiple metrics (OEC)	Makes it harder to game one number
Set guardrails	Tracks unintended side effects like churn.
Audit the process	Look for suspicious last setups or exclusions.
Use blind experimentation	Don't let teams know when group is test/control.
Reward long-term impact.	Prevent short-term metric inflation.

(3) The Lucas Critique.

(1) What is the Lucas Critique?

→ "You can't reliably use historical models to predict the effect of a policy change"

→ Simply,
"You can't predict the effect of new policy using old data – because people change their behavior when the rules changes."

This principle comes from economist Robert Lucas, originally about macroeconomic policy, but it widely relevant in experiments, A/B testing and product metrics.

(2) Real-World Analogy and Example

You run a streaming platform
You've modeled this from historical data:

$$\text{Revenue} = f(\text{Ads}) = \text{Number of Ads} \times \text{Click-Through Rate (CTR)}$$

Let's say:
- You show 2 ads per user now.
- You estimate that adding 1 ad will give ~~you~~ 10% more revenue.

So, you increase ads from 2 → 3

But what actually happens?

↳ Users get annoyed.
↳ CTR drops
↳ Some quit the platform

So your original prediction was wrong
why?
Because your model assumed user behavior would stay fixed — but it changed once you changed the policy.

This is the Lucas Critique.

Pure Math Example

Let's formalize

$$\text{Revenue} = \text{Ads} \times \text{CTR}$$

Now suppose

- CTR = 0.05 when showing 2 ads
- So current revenue per user = $2 \times 0.05 = 0.10$.

You think:

Let's add 1 more ad ⇒ 3 ads
Now revenue = $3 \times 0.05 = 0.15$

But after the change.

- users are frustrated.
- They stop clicking
- Now CTR = 0.03

So actual revenue:

$$\text{Revenue} = 3 \times 0.03 = 0.09$$

You predicted an increase to 0.15
but instead you lost revenue!

(3) Why this happens (Lucas Critique in formulas)

You assumed:

$$CTR = \text{constant}$$

But in reality

$$CTR = g(Ads) = \text{a decreasing function}$$

So the true model is:

$$Revenue = Ads \times g(Ads)$$

And when you change Ads, the online user behavior function $g(\cdot)$ shift.

That's what Lucas warned:

> "You cannot use a model with fixed behavior to predict the impact of a policy that changes incentives!".

(4) Overlap with A/B testing

This is why short-term A/B test result may mislead:

- You test new pricing
- In the short term, revenue increases.
- But in the long term
  - a) users adapt
  - b) They spend less, churn more, or find alternatives

Hence: don't overemphasize blindly from test to reality.